

- 2 A student adds excess large pieces of magnesium carbonate, MgCO_3 , to dilute hydrochloric acid, HCl , and measures the volume of carbon dioxide gas, CO_2 , given off.

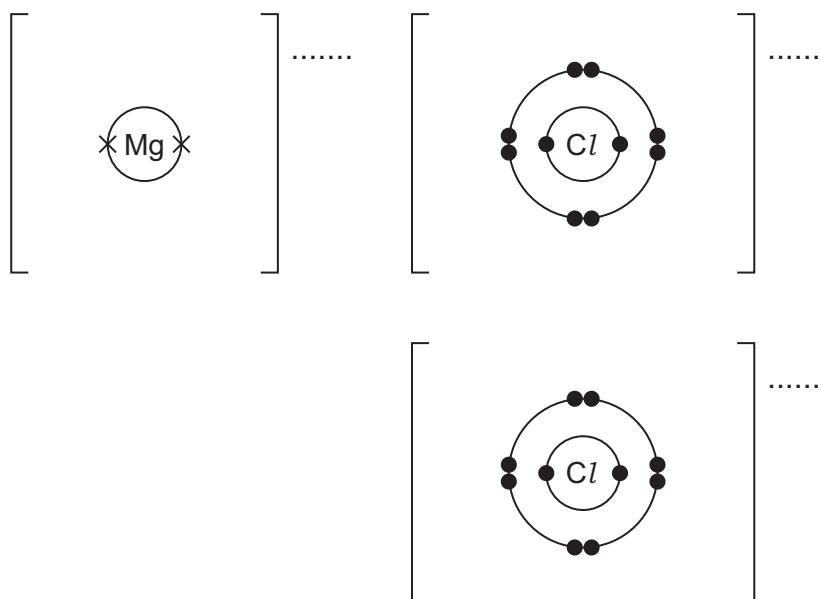
(a) Add the missing state symbols to the chemical equation for the reaction.



- (b) Complete the dot-and-cross diagram to show the electron arrangement of the ions in magnesium chloride.

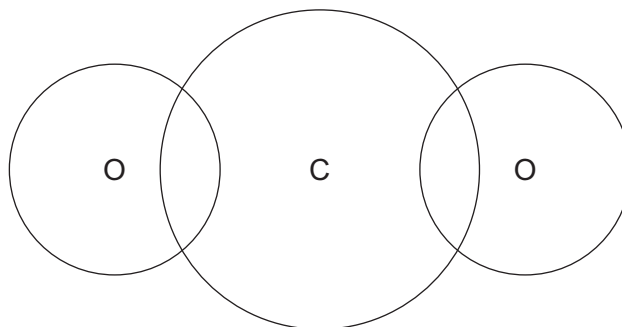
The inner shells have been drawn.

Give the charges on the ions.



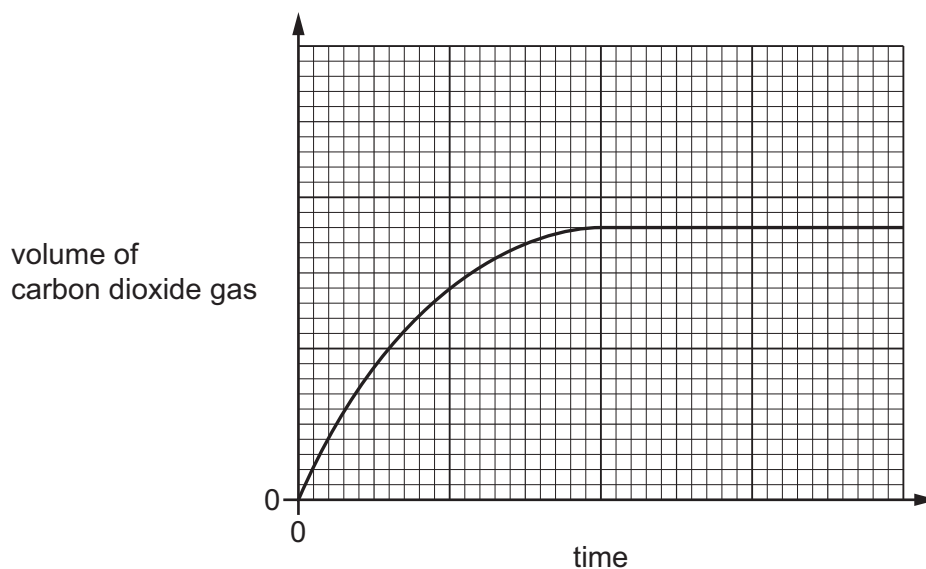
[3]

- (c) Complete the dot-and-cross diagram to show the electron arrangement in a molecule of carbon dioxide.
Show outer shell electrons only.



[2]

(d) The graph shows how the volume of carbon dioxide gas changes with time.



(i) Describe how the graph shows that the rate of this reaction decreases as time increases.

..... [1]

(ii) Explain, in terms of particles, why the rate of this reaction decreases as time increases.

..... [2]

(iii) The student repeats the experiment using powdered MgCO_3 instead of large pieces.

All other conditions stay the same.

On the grid, draw the line expected when powdered MgCO_3 is used instead of large pieces. [2]

- (e) Determine the volume of CO_2 gas given off when excess MgCO_3 is added to 25.0 cm^3 of 0.400 mol/dm^3 HCl at room temperature and pressure.



Use the following steps.

- Calculate the number of moles of HCl in 25.0 cm^3 of 0.400 mol/dm^3 of acid.

..... mol

- Determine the number of moles of CO_2 gas given off.

..... mol

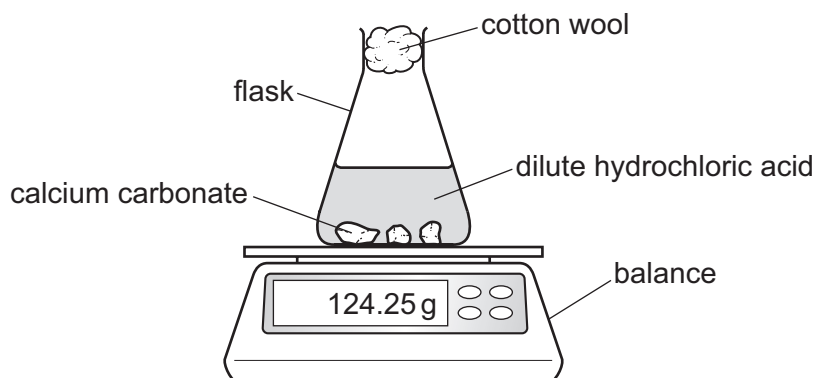
- Calculate the volume of CO_2 gas given off in cm^3 .

..... cm^3
[3]

[Total: 14]

- 1 Calcium carbonate reacts with dilute hydrochloric acid. The products of the reaction are aqueous calcium chloride, water and carbon dioxide gas.

A student investigated the rate of the reaction between calcium carbonate and dilute hydrochloric acid using the apparatus shown.



The mass of the flask and contents was recorded every 30 seconds.
When the reaction stopped there were still small pieces of calcium carbonate in the flask.

- (a) State what happens to the reading on the balance as the reaction takes place.
Explain your answer.

reading on balance

explanation

[2]

- (b) There is a piece of cotton wool in the neck of the flask.

- (i) Suggest why a bung is **not** used in the neck of the flask.

.....

..... [1]

- (ii) Suggest why cotton wool is placed in the neck of the flask rather than leaving the flask open.

.....

..... [1]

- (c) State which reactant is in excess.

..... [1]

- (d) Describe how crystals of calcium chloride can be obtained from the mixture left in the flask after the reaction has stopped.

.....

.....

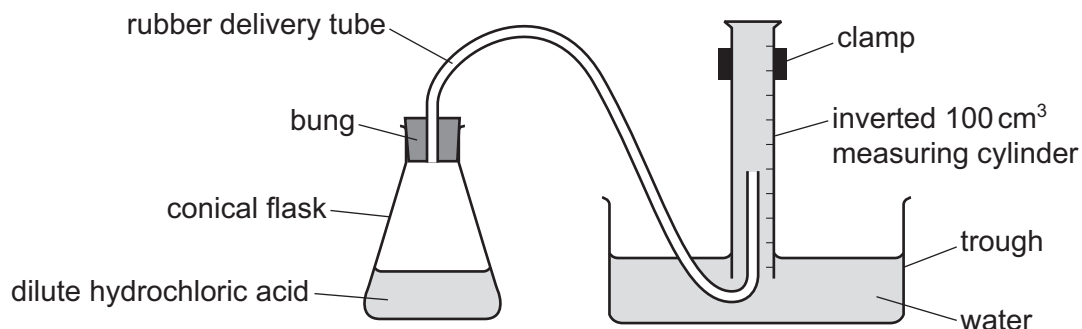
.....

..... [3]

[Total: 8]

- 2 A student investigated the rate at which hydrogen gas is made when magnesium reacts with two different solutions of dilute hydrochloric acid, **C** and **D**, with different concentrations. The dilute hydrochloric acid was in excess in both experiments.

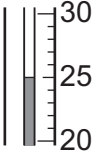
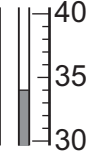
Two experiments were done using the apparatus shown.

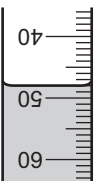
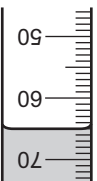
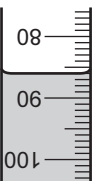
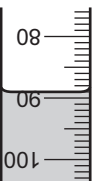
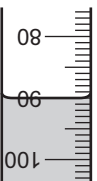
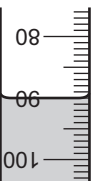


Experiment 1

- A measuring cylinder was used to pour 50 cm³ of dilute hydrochloric acid **C** into a conical flask.
- The initial temperature of the dilute hydrochloric acid was measured using a thermometer.
- The apparatus was set up as shown in the diagram.
- The bung was removed from the conical flask and a coiled 5 cm length of magnesium ribbon was added to the flask. The bung was replaced immediately and a timer started.
- The volume of gas collected in the inverted measuring cylinder was recorded every 20 seconds for 160 seconds.
- The final temperature of the dilute hydrochloric acid in the flask was measured using a thermometer.

- (a) Use the thermometer diagrams and the diagrams of inverted measuring cylinders to complete the tables.

initial		final	
thermometer diagram	temperature / °C	thermometer diagram	temperature / °C
			

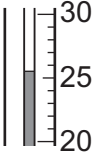
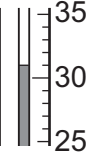
time / s	20	40	60	80	100	120	140	160
diagrams of inverted measuring cylinder								
volume of gas collected / cm ³								

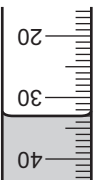
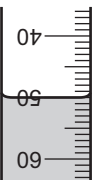
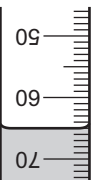
[2]

(b) *Experiment 2*

- Experiment 1 was repeated using 50 cm³ of dilute hydrochloric acid **D** instead of dilute hydrochloric acid **C**.

Use the thermometer diagrams and the diagrams of inverted measuring cylinders to complete the tables.

initial		final	
thermometer diagram	temperature / °C	thermometer diagram	temperature / °C
			

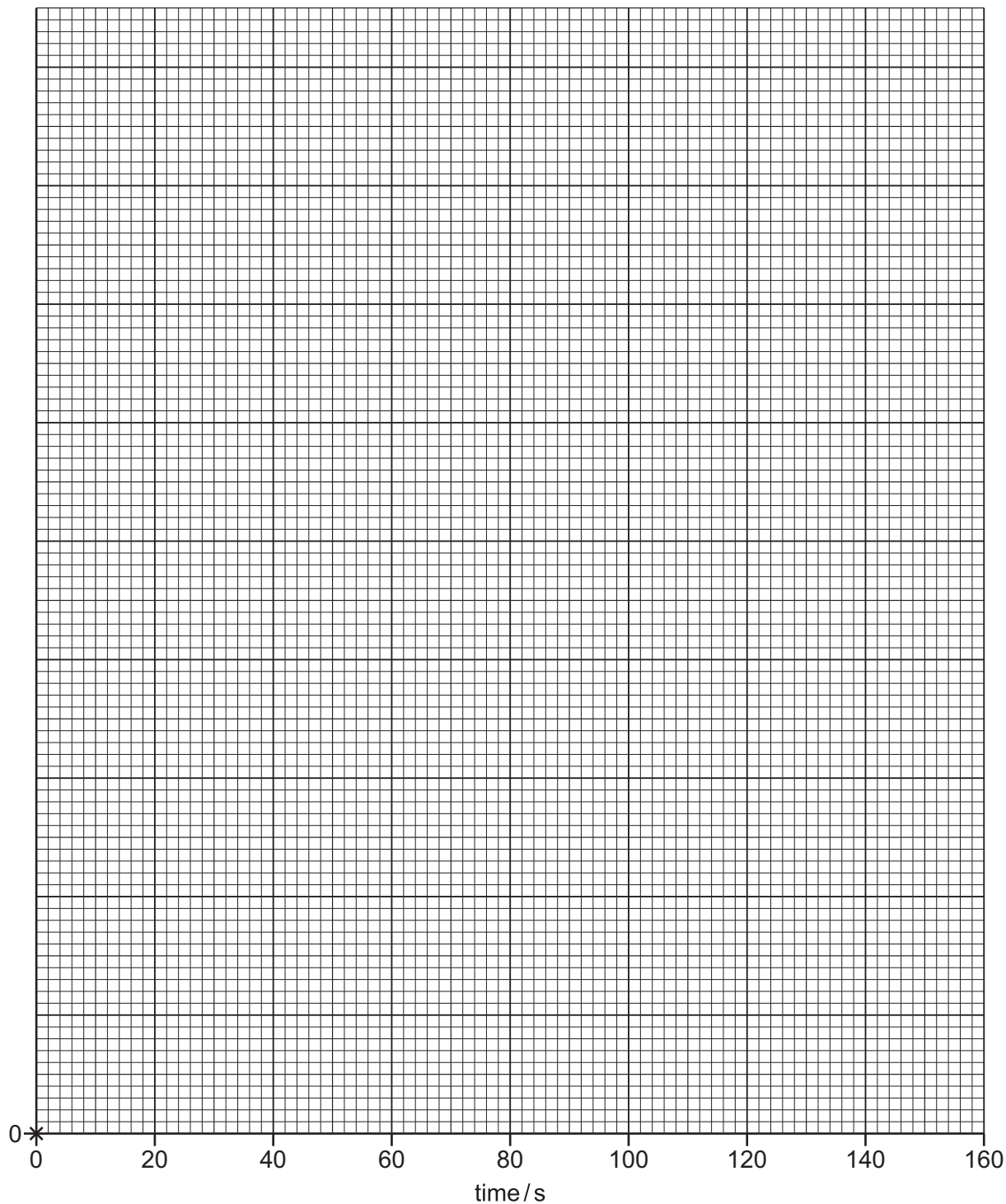
time / s	20	40	60	80	100	120	140	160
diagrams of inverted measuring cylinder								
volume of gas collected / cm ³								

[3]

- (c) Complete a suitable scale on the y -axis and plot your results from Experiments 1 and 2 on the grid.

Draw **two** smooth line graphs. The lines must pass through (0,0). Clearly label your lines.

volume of
gas collected
/cm³



[5]

- (d) **From your graph**, deduce the volume of gas that was collected after 50 seconds in Experiment 2.

Show clearly **on the grid** how you worked out your answer.

volume of gas = [3]

- (e) Explain what can be deduced about the concentrations of dilute hydrochloric acid **C** and dilute hydrochloric acid **D**.

.....
.....
.....
..... [2]

- (f) (i) State what happens to the temperature of the dilute hydrochloric acid during Experiment 1.

..... [1]

- (ii) State what effect this temperature change has on the total volume of gas made when the reaction has finished.

..... [1]

- (iii) Describe a change that can be made to the apparatus or reagents to reduce the temperature change of the acid in Experiment 1.

..... [1]

- (g) Suggest why it is important to replace the bung in the conical flask immediately after adding the magnesium ribbon.

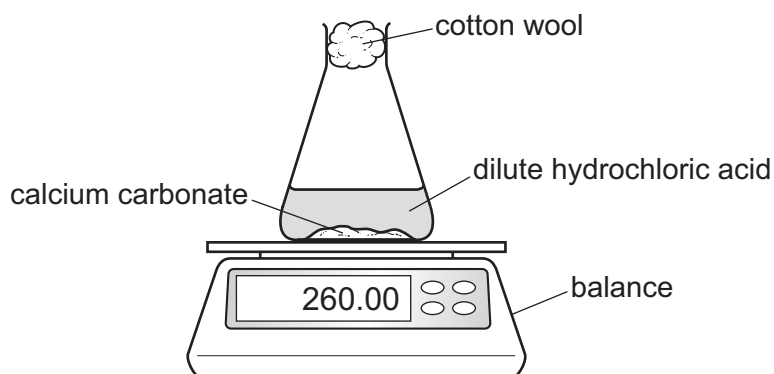
.....
..... [1]

- (h) State the advantage of measuring the volume of gas collected every 10 seconds rather than every 20 seconds.

..... [1]

[Total: 20]

- 4 Dilute hydrochloric acid reacts with calcium carbonate to make carbon dioxide gas. The apparatus shown in the diagram can be used to follow the progress of the reaction. The carbon dioxide gas leaves the flask causing the mass shown on the balance to decrease.



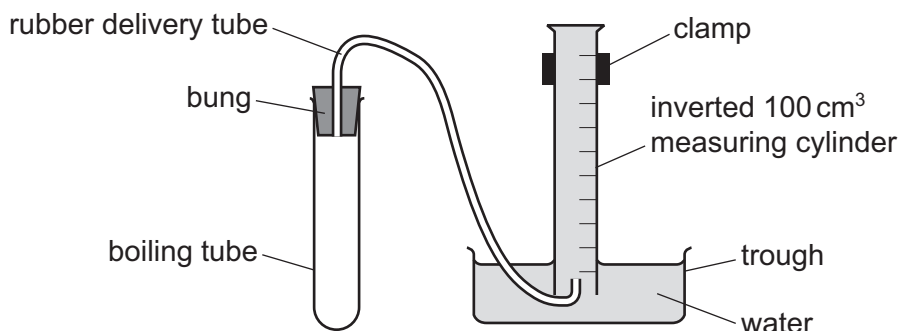
Plan an investigation, using the apparatus shown in the diagram, to find out how the temperature of the dilute hydrochloric acid affects the rate of the reaction. Your plan should include how your results will show how the temperature of the dilute hydrochloric acid affects the rate of the reaction.

You are provided with dilute hydrochloric acid, calcium carbonate and common laboratory apparatus.

[6]

- 2 A student investigated the volume of gas made when sodium carbonate reacts with dilute hydrochloric acid.

Five experiments were carried out using the apparatus shown.



Experiment 1

- Using a measuring cylinder, 16 cm³ of dilute hydrochloric acid was poured into a boiling tube.
- The apparatus was set up as shown in the diagram.
- The bung was removed from the boiling tube.
- 2.5 g of sodium carbonate was added to the boiling tube and the bung was immediately replaced.
- When no more gas was being collected, the volume of gas in the measuring cylinder was measured.

Experiment 2

- Experiment 1 was repeated using 14 cm³ of dilute hydrochloric acid instead of 16 cm³.

Experiment 3

- Experiment 2 was repeated using 12 cm³ of dilute hydrochloric acid instead of 14 cm³.

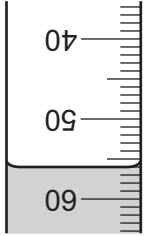
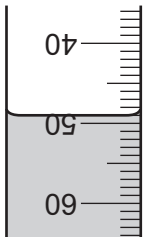
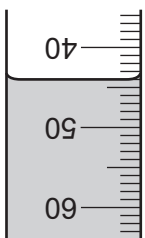
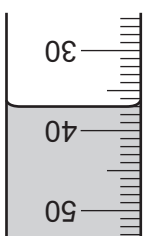
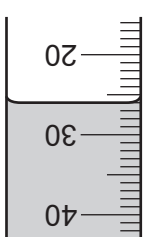
Experiment 4

- Experiment 3 was repeated using 10 cm³ of dilute hydrochloric acid instead of 12 cm³.

Experiment 5

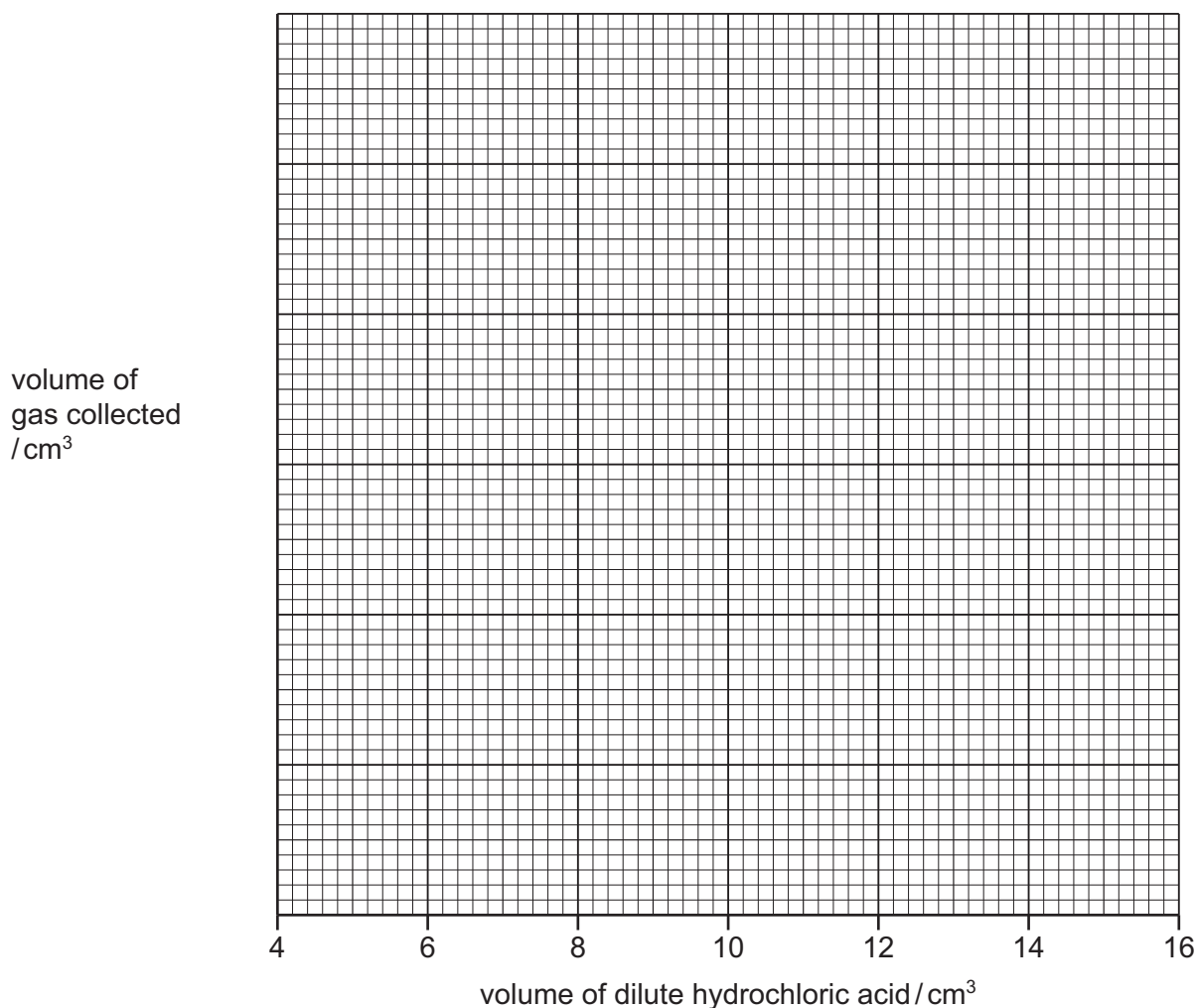
- Experiment 4 was repeated using 6 cm³ of dilute hydrochloric acid instead of 10 cm³.

- (a) Use the information in the description of the experiments and the inverted measuring cylinder diagrams to complete the table.

experiment	volume of dilute hydrochloric acid / cm ³	inverted measuring cylinder diagram	volume of gas collected / cm ³
1			
2			
3			
4			
5			

[3]

- (b) Write a suitable scale on the y-axis and plot the results from Experiments 1 to 5 on the grid. Draw a straight line of best fit.



[4]

- (c) (i) **From your graph**, deduce the volume of gas that would be collected if 7 cm³ of dilute hydrochloric acid was used.

Show clearly **on the grid** how you worked out your answer.

..... cm³
[2]

- (ii) The volume of gas made per cm³ of dilute hydrochloric acid can be calculated using the equation shown.

$$\text{volume of gas per cm}^3 \text{ of acid} = \frac{\text{volume of gas collected in cm}^3}{\text{volume of acid in cm}^3}$$

Use this equation and your answer to (c)(i) to calculate the volume of gas made per cm³ of dilute hydrochloric acid.

..... [1]

(d) The bung was removed and then replaced immediately after the sodium carbonate was added to the boiling tube.

(i) Explain why the bung must be replaced immediately after the sodium carbonate is added to the boiling tube.

.....
..... [1]

(ii) Explain how the apparatus could be altered so that the bung does **not** have to be removed. You may draw a diagram to explain your answer.

.....
..... [2]

(e) State **one** advantage of using a burette rather than a measuring cylinder to measure the volume of the dilute hydrochloric acid.

..... [1]

(f) In Experiments 1 to 5, the sodium carbonate was in excess.

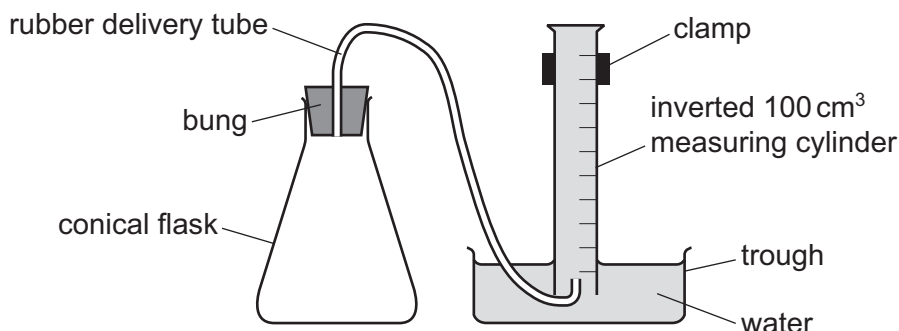
Sketch **on the grid** the graph you would expect if all of the experiments were repeated using dilute hydrochloric acid of half the concentration.

Label your line **F**. [2]

[Total: 16]

- 2 A student investigated the rate at which hydrogen gas was made when magnesium reacted with dilute sulfuric acid.

Five experiments were carried out using the apparatus shown.



Experiment 1

- Using a measuring cylinder, 25 cm³ of dilute sulfuric acid was poured into a conical flask.
- Using a different measuring cylinder, 30 cm³ of distilled water was poured into the conical flask.
- The apparatus was set up as shown in the diagram.
- The bung was removed from the conical flask.
- A coiled length of magnesium ribbon was added to the conical flask, the bung was replaced immediately and a timer started.
- The volume of gas collected in the inverted measuring cylinder after 30 seconds was measured.

Experiment 2

- Experiment 1 was repeated using 20 cm³ of distilled water instead of 30 cm³.

Experiment 3

- Experiment 1 was repeated using 10 cm³ of distilled water instead of 30 cm³.

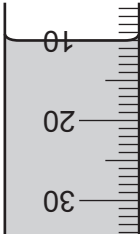
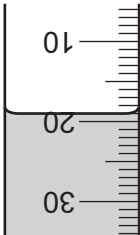
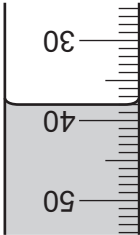
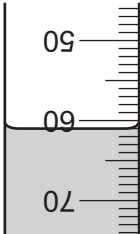
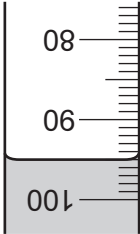
Experiment 4

- Experiment 1 was repeated using 5 cm³ of distilled water instead of 30 cm³.

Experiment 5

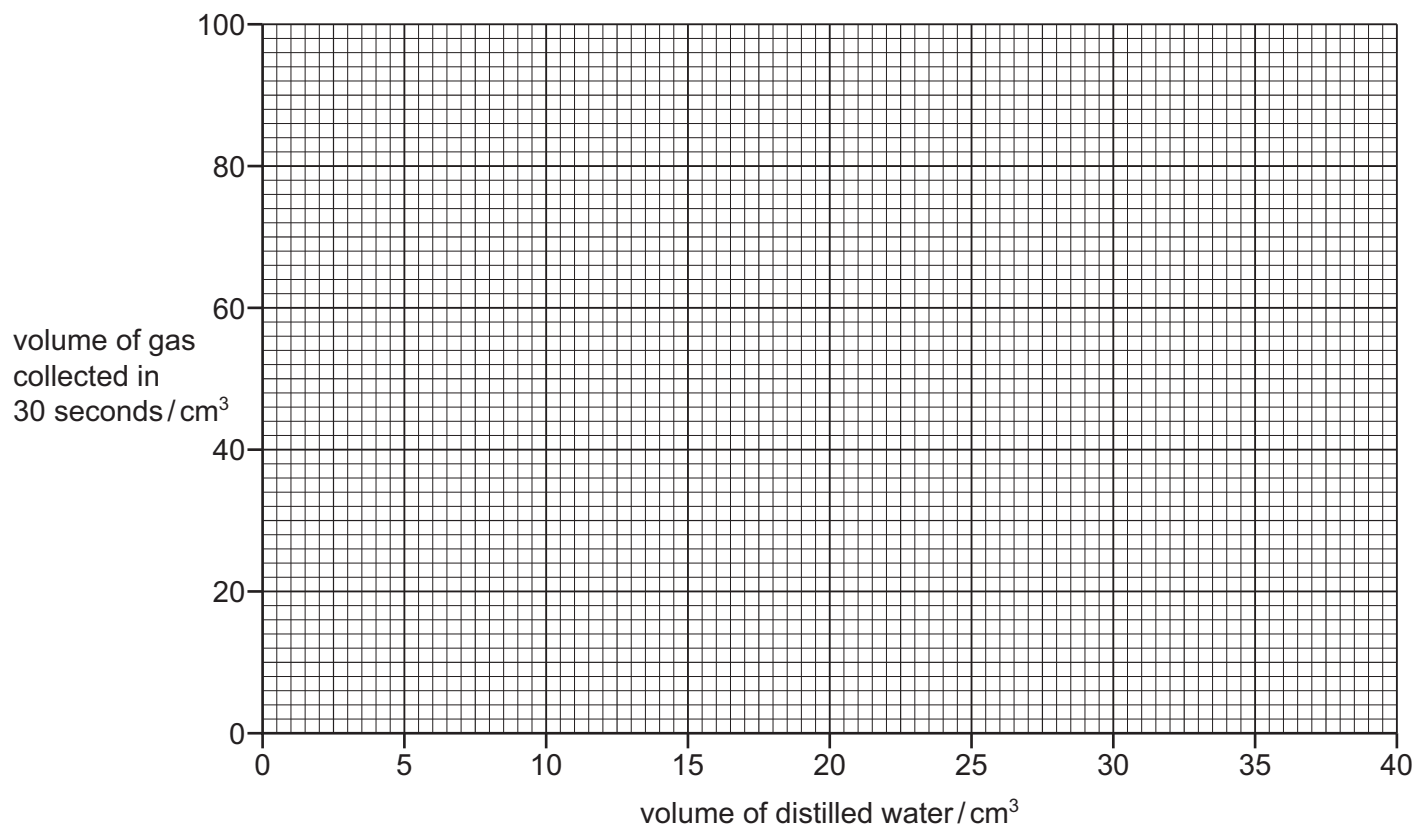
- Experiment 1 was repeated without adding any distilled water to the dilute sulfuric acid.

(a) Use the information in the description of the experiments and the inverted measuring cylinder diagrams to complete the table.

experiment	volume of dilute sulfuric acid / cm ³	volume of distilled water / cm ³	inverted measuring cylinder diagram	volume of gas collected in 30 seconds / cm ³
1				
2				
3				
4				
5				

[4]

(b) Plot the results from Experiments 1 to 5 on the grid. Draw a smooth curve of best fit.



[3]

(c) Extrapolate (extend) the line on your graph and deduce the volume of gas that would be collected in 30 seconds if 35 cm³ of distilled water was added to the dilute sulfuric acid.

..... cm³
[2]

- (d) The rate of reaction can be calculated using the equation shown.

$$\text{rate of reaction} = \frac{\text{volume of gas collected}}{\text{time taken to collect the gas}}$$

- (i) Use this equation to calculate the rate of reaction in Experiment 3. Give the units for the rate you have calculated.

rate =

units =

[2]

- (ii) State which Experiment, 1, 2, 3, 4 or 5, had the highest rate of reaction.

..... [1]

- (e) The volume of the dilute sulfuric acid was measured using a measuring cylinder. A 25 cm³ pipette could have been used instead of a measuring cylinder.

- (i) State **one** advantage of using a 25 cm³ pipette instead of a measuring cylinder.

..... [1]

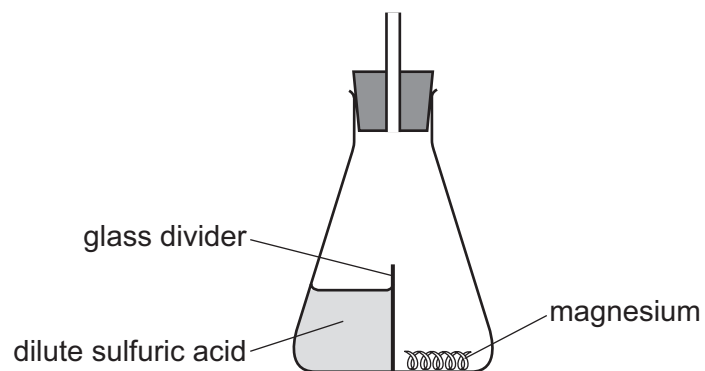
- (ii) State **one** disadvantage of using a 25 cm³ pipette instead of a measuring cylinder.

..... [1]

- (f) Name another item of apparatus, which can be used instead of an inverted measuring cylinder, to collect and measure the volume of gas made in the reaction.

..... [1]

(g) The diagram shows a modified conical flask that could be used in this investigation.



Explain the advantage of using this type of conical flask instead of the type used in the investigation.

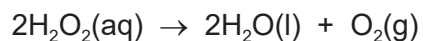
.....

.....

..... [2]

[Total: 17]

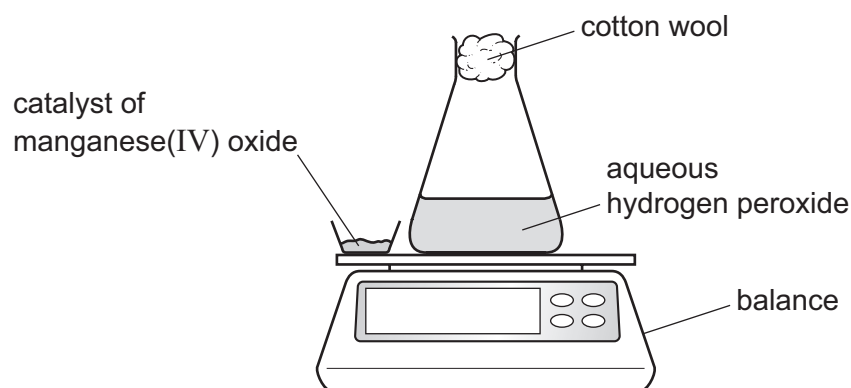
- 6 A student investigates the decomposition of hydrogen peroxide in the presence of a catalyst of manganese(IV) oxide.



- (a) State the meaning of the term *catalyst*.

.....
..... [2]

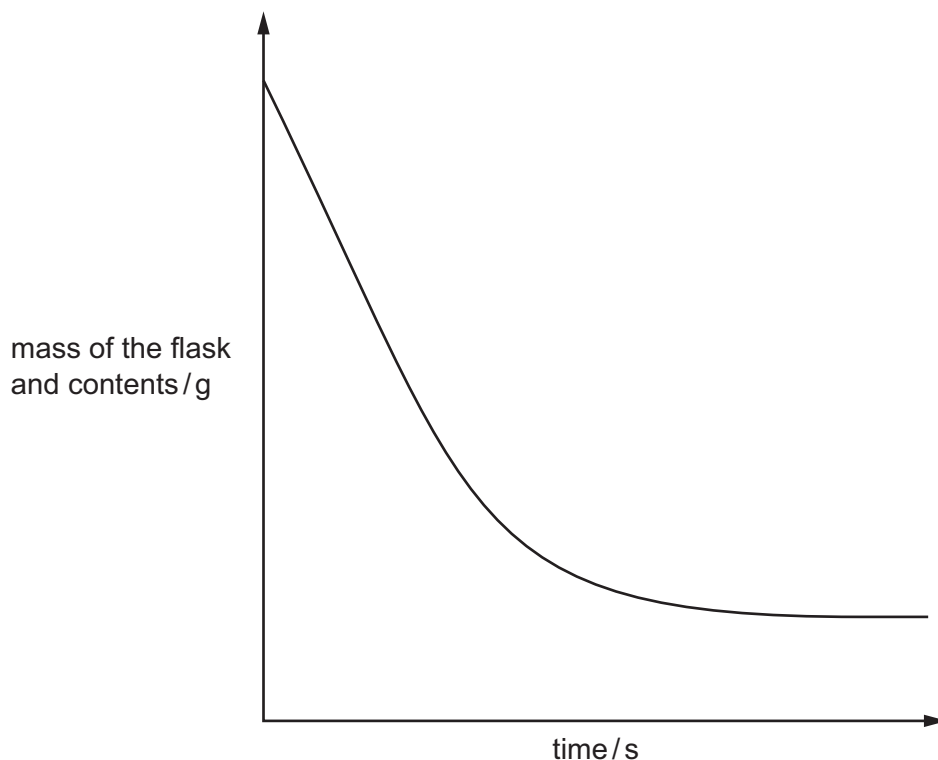
- (b) The diagram shows the equipment the student uses.



The student uses this method:

- the catalyst is added to the aqueous hydrogen peroxide
- the stop-clock is started
- the mass of the flask and contents is recorded at regular time intervals.

A graph of the mass of the flask and contents against time is shown.



- (i) Suggest why the mass of the flask and contents decreases as time increases.

..... [1]

- (ii) Describe what happens to the rate of the reaction as time increases.

.....

..... [2]

- (c) The student repeats the experiment at a higher temperature. All other conditions stay the same. The rate of reaction increases.

- (i) Explain, in terms of collisions between particles, why the rate of reaction increases at a higher temperature.

.....

.....

.....

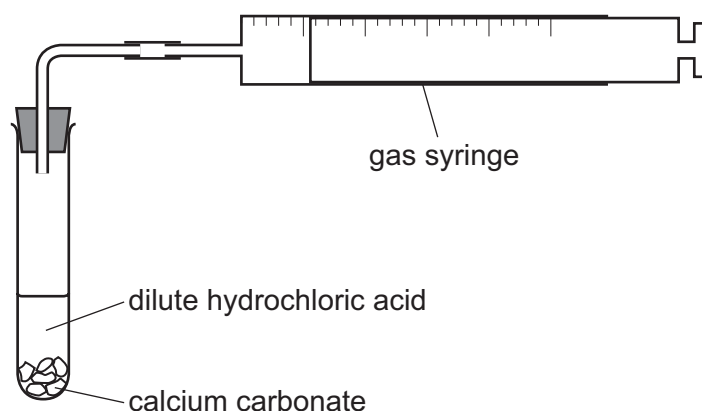
..... [3]

- (ii) Draw a line on the graph in (b) for the experiment at a higher temperature. [2]

[Total: 10]

- 2 A student investigated the rate of reaction between small lumps of calcium carbonate and dilute hydrochloric acid.

The experiment was done at two different temperatures using the apparatus shown in the diagram. All other conditions were kept the same.



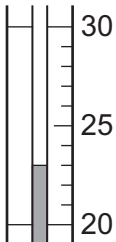
Experiment 1

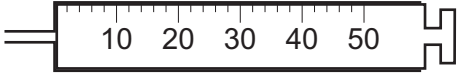
- Using a 50 cm³ measuring cylinder, 25 cm³ of dilute hydrochloric acid was poured into a boiling tube.
- The temperature of the dilute hydrochloric acid was measured using a thermometer.
- 10 g of lumps of calcium carbonate were added to the boiling tube, the bung replaced and the stop-watch started.
- The volume of gas collected in the gas syringe was measured every 50 seconds for 250 seconds.

Experiment 2

- Using a 50 cm³ measuring cylinder, 25 cm³ of dilute hydrochloric acid was poured into a boiling tube.
- The dilute hydrochloric acid in the boiling tube was warmed using a Bunsen burner.
- The temperature of the dilute hydrochloric acid was measured using a thermometer.
- 10 g of lumps of calcium carbonate were added to the boiling tube, the bung replaced and the stop-watch started.
- The volume of gas collected in the gas syringe was measured every 50 seconds for 250 seconds.

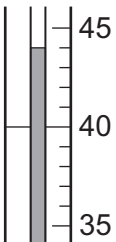
(a) Use the diagrams to complete the tables for Experiment 1.

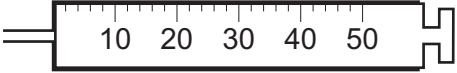
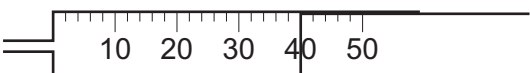
thermometer diagram	temperature of dilute hydrochloric acid/°C
	

time/s	gas syringe diagram	volume of gas collected/cm ³
0		
50		
100		
150		
200		
250		

[2]

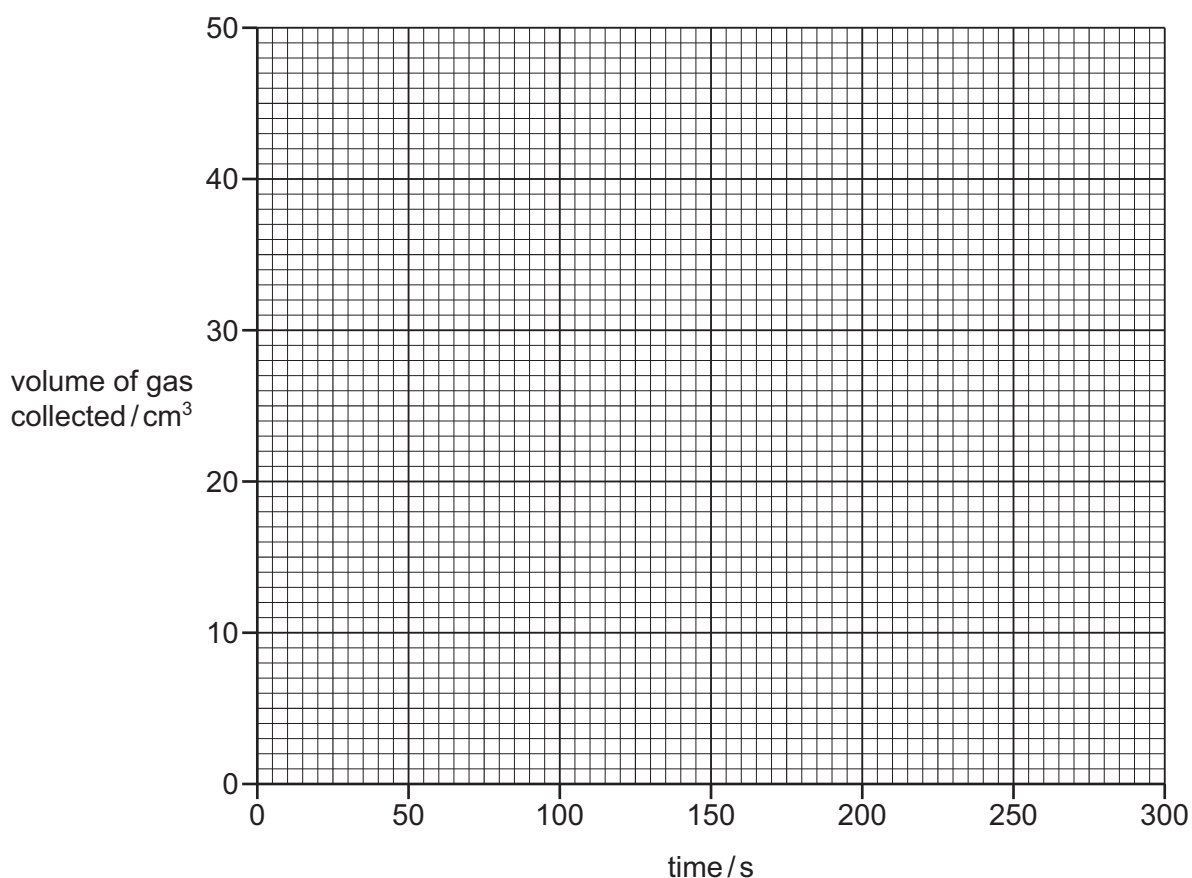
(b) Use the diagrams to complete the tables for Experiment 2.

thermometer diagram	temperature of dilute hydrochloric acid / °C
	

time / s	gas syringe diagram	volume of gas collected / cm ³
0		
50		
100		
150		
200		
250		

[2]

- (c) Plot the results from Experiment 1 and Experiment 2 on the grid. Draw a curve of best fit for each experiment. Label your curves.



[5]

- (d) **From your graph**, deduce the volume of gas collected in Experiment 2 after 120 seconds.

Show clearly **on the grid** how you worked out your answer.

..... cm^3
[2]

- (e) Explain how the results show that the reaction in Experiment 2 has stopped.

.....
..... [1]

- (f) Predict the volume of gas that would be collected in Experiment 1 after 800 seconds. Explain your answer.

volume of gas collected after 800 seconds cm^3

explanation

.....
[2]

- (g) A student stated it would be an improvement to measure the volume of gas collected every 25 seconds.

Explain why this is an improvement.

.....
.....
..... [2]

- (h) State **two** changes to the apparatus to improve the accuracy of the results obtained if the experiment is repeated using the same thermometer.

change 1
.....
change 2
..... [2]

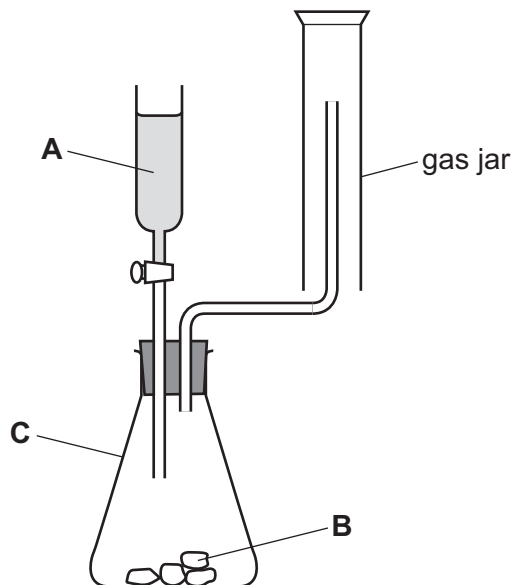
- (i) Describe how the method used in Experiment 2 could be changed so that results can be obtained using dilute hydrochloric acid at 1 °C.

.....
..... [1]

[Total: 19]

- 1 Carbon dioxide is a colourless gas that is denser than air. Carbon dioxide can be made by reacting marble chips with dilute hydrochloric acid.

A student tried to make and collect carbon dioxide gas using the apparatus shown.



- (a) (i) Name the substances labelled **A** and **B**.

A

B [1]

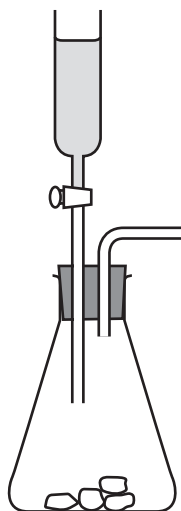
- (ii) Name the item of apparatus labelled **C**.

..... [1]

- (b) Explain why very little carbon dioxide gas would be collected using the apparatus shown.

.....
..... [2]

- (c) Complete the diagram to show how carbon dioxide gas could be collected and the volume measured.



[2]

- (d) At the end of the experiment there were unreacted marble chips and aqueous calcium chloride in the item of apparatus labelled **C**.

Describe how you would find the mass of unreacted marble chips in apparatus **C**.

.....

.....

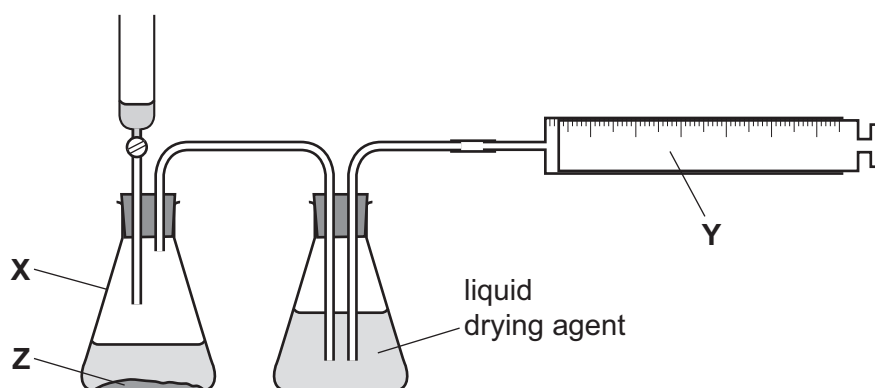
.....

..... [3]

[Total: 9]

- 1 Hot concentrated hydrochloric acid reacts with solid manganese(IV) oxide to make chlorine gas. Chlorine gas can be dried by bubbling it through a liquid drying agent.

The diagram shows the apparatus used to make and collect a sample of dry chlorine gas. There is one error in the diagram.



- (a) Name the items of apparatus labelled X and Y.

X
Y [2]

- (b) Name the substance labelled Z.

..... [1]

- (c) On the diagram draw one arrow to show where heat should be applied so that chlorine gas is made. [1]

- (d) There is one error in the way the apparatus has been set up.

- (i) On the diagram draw a circle around the error in the apparatus. [1]

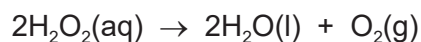
- (ii) Describe what would happen if the apparatus is used before the error is corrected.

.....
..... [1]

[Total: 6]

- 4 Catalysts are substances which increase the rate of a reaction but are unchanged at the end of the reaction.

Aqueous hydrogen peroxide decomposes slowly to form water and oxygen.



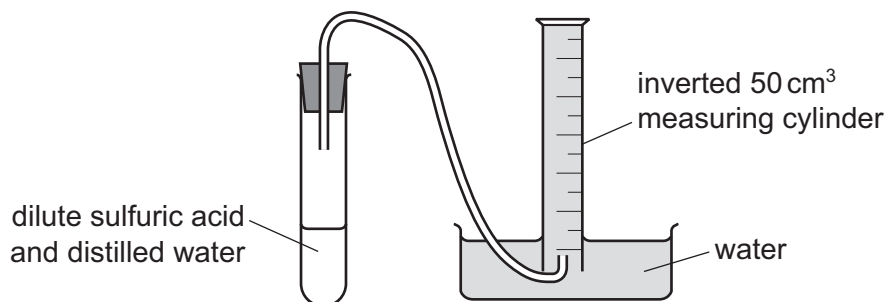
Copper(II) oxide is an insoluble solid.

Plan an investigation to find out if copper(II) oxide is a catalyst for the decomposition of hydrogen peroxide. You must include how your results will tell you if copper(II) oxide is a catalyst. You have access to copper(II) oxide, aqueous hydrogen peroxide and all normal laboratory apparatus.

[6]

- 2 A student investigated the time taken to collect 40 cm^3 of hydrogen gas when magnesium reacts with dilute sulfuric acid.

Five experiments were done using the apparatus shown.



Experiment 1

- Using a measuring cylinder, 8 cm^3 of dilute sulfuric acid was poured into the boiling tube.
- Using a second measuring cylinder, 12 cm^3 of distilled water was added to the acid in the boiling tube.
- The apparatus was set up as shown in the diagram, ensuring the inverted measuring cylinder was full of water.
- The bung was removed from the boiling tube.
- A coiled length of magnesium ribbon was added to the boiling tube, the bung was immediately replaced and a timer started.
- The time taken for 40 cm^3 of gas to be collected was measured.
- The student felt the outside of the boiling tube.

- (a) (i) The student noticed that the boiling tube was warm.

What does this tell you about the type of reaction?

..... [1]

- (ii) Describe **one** change that could be made to the apparatus to help keep the temperature of the contents of the boiling tube constant during the reaction.

..... [1]

Experiment 2

- The boiling tube was rinsed out with distilled water.
- Experiment 1 was repeated using 10 cm^3 of dilute sulfuric acid and 10 cm^3 of distilled water.

Experiment 3

- Experiment 2 was repeated using 12 cm^3 of dilute sulfuric acid and 8 cm^3 of distilled water.

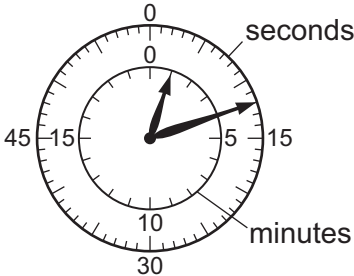
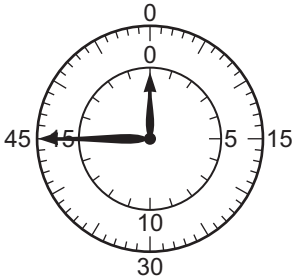
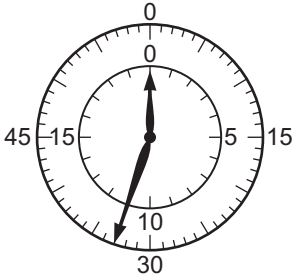
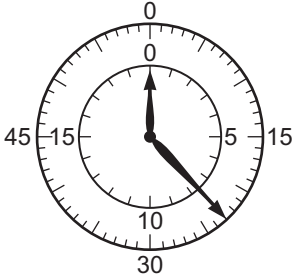
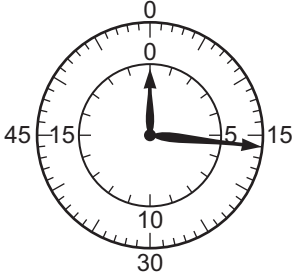
Experiment 4

- Experiment 2 was repeated using 16 cm^3 of dilute sulfuric acid and 4 cm^3 of distilled water.

Experiment 5

- Experiment 2 was repeated using 20 cm^3 of dilute sulfuric acid and no distilled water.

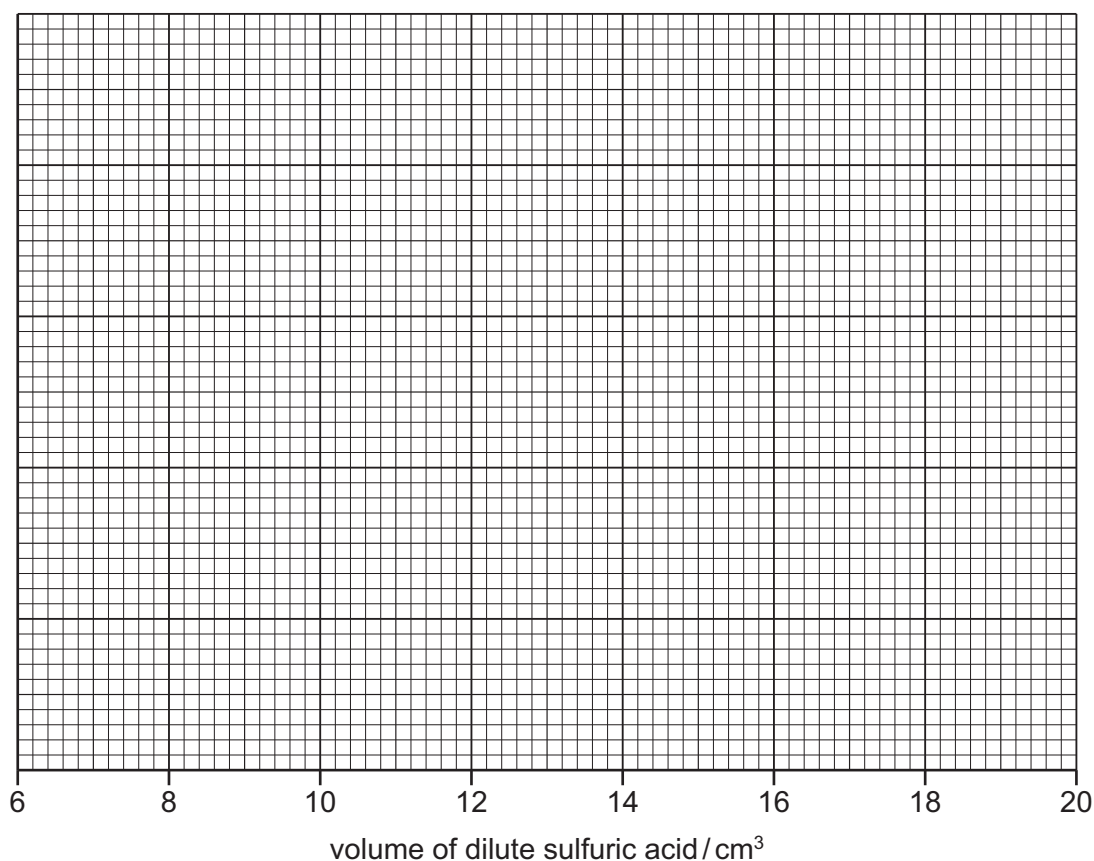
(b) Use the information in the description of the experiments and the timer diagrams to complete the table. Record the time in **seconds**.

experiment	volume of dilute sulfuric acid / cm ³	volume of distilled water / cm ³	timer diagram	time to collect 40 cm ³ of gas/s
1	8			
2	10			
3	12			
4	16			
5	20			

[4]

- (c) Add a suitable scale to the y-axis and plot the results from Experiments 1 to 5 on the grid. Draw a smooth line graph.

time to collect
40 cm³ of gas / s



[4]

- (d) (i) **From your graph**, deduce the time taken to collect 40 cm³ of gas if the experiment was repeated using 9 cm³ of dilute sulfuric acid.

Show clearly **on the grid** how you worked out your answer.

..... s
[2]

- (ii) What volume of distilled water would be needed if the experiment was repeated using 9 cm³ of dilute sulfuric acid?

..... cm³ [1]

- (e) The rate of reaction can be calculated using the equation shown.

$$\text{rate of reaction} = \frac{\text{volume of gas collected}}{\text{time taken to collect the gas}}$$

- (i) Use this equation to calculate the rate of reaction in Experiment 1. Give the units for the rate of reaction you have calculated.

rate of reaction = units = [2]

- (ii) In which Experiment, 1, 2, 3, 4 or 5, was the rate of reaction greatest?

..... [1]

- (f) Why would measuring the volume of dilute sulfuric acid with a burette rather than a measuring cylinder be an improvement?

.....
..... [1]

- (g) The magnesium starts to react with the dilute sulfuric acid as soon as it is added.

- (i) Why does this decrease the accuracy of the investigation?

..... [1]

- (ii) Describe **one** improvement that you could make to overcome this problem.

.....
.....
..... [1]

[Total: 19]

- 4 Oxygen is produced by the decomposition of hydrogen peroxide. Manganese(IV) oxide is the catalyst for this reaction.

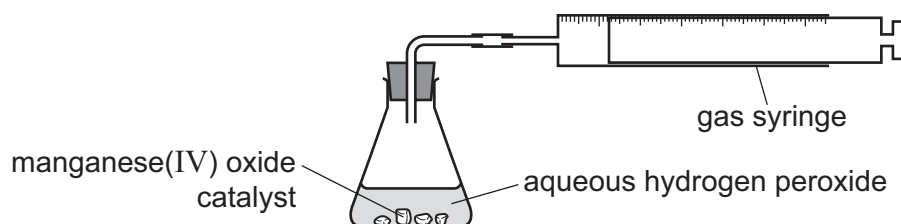
(a) What is meant by the term *catalyst*?

.....

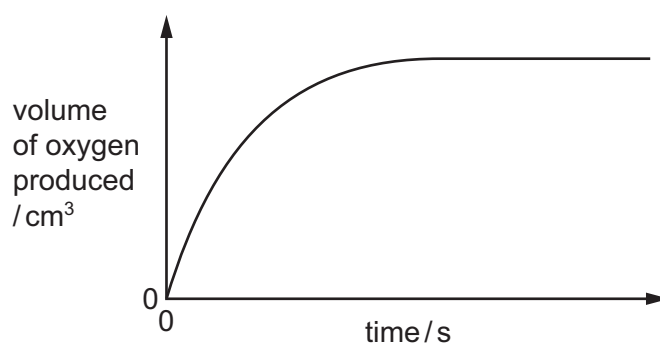
.....

..... [2]

- (b) A student measures the volume of oxygen produced at regular time intervals using the apparatus shown. Large lumps of manganese(IV) oxide are used.



A graph of the results is shown.



What happens to the **rate** of this reaction as time increases?
In your answer, explain why the rate changes in this way.

.....

.....

.....

.....

..... [4]

- (c) The experiment is repeated using the same mass of manganese(IV) oxide. Powdered manganese(IV) oxide is used instead of large lumps. All other conditions stay the same.

Sketch a graph on the axes in (b) to show how the volume of oxygen changes with time. [2]

- (d) In terms of particles, explain what happens to the rate of this reaction when the temperature is increased.

.....

.....

.....

.....

..... [3]

- (e) The equation for the decomposition of hydrogen peroxide is shown.



25.0 cm³ of aqueous hydrogen peroxide forms 48.0 cm³ of oxygen at room temperature and pressure (r.t.p.).

Calculate the concentration of aqueous hydrogen peroxide at the start of the experiment using the following steps.

- Calculate the number of moles of oxygen formed.

..... mol

- Deduce the number of moles of hydrogen peroxide that decomposed.

..... mol

- Calculate the concentration of hydrogen peroxide in mol/dm³.

..... mol/dm³
[3]

- (f) Oxygen can also be produced by the decomposition of potassium chlorate(V), KClO₃.

The only products of this decomposition are potassium chloride and oxygen.

Write a chemical equation for this decomposition.

..... [2]

[Total: 16]

- 4 Cobalt, manganese and nickel are metals. They react with dilute hydrochloric acid to form hydrogen gas.

Plan an investigation to find the order of reactivity of these three metals.

You are provided with:

- samples of each metal
- dilute hydrochloric acid
- common laboratory apparatus.

Your plan must make it clear how your investigation will be a fair test and how you will use your results to place the metals in order of reactivity.

..... [6]

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- 2 A student investigated the rate of a reaction between sodium metabisulfite and potassium iodate. In the reaction, starch was used as an indicator. At first the reacting mixture remained colourless but then suddenly changed to a blue-black colour.

Five experiments were done. In each experiment the total volume of liquid was 45 cm^3 .

Experiment 1

- Using a 10 cm^3 measuring cylinder, 5 cm^3 of aqueous sodium metabisulfite was poured into a beaker.
- Using another 10 cm^3 measuring cylinder, 5 cm^3 of aqueous starch was poured into the beaker.
- Using a 25 cm^3 measuring cylinder, 15 cm^3 of distilled water was poured into the beaker.
- Using another 25 cm^3 measuring cylinder, 20 cm^3 of aqueous potassium iodate was poured into the beaker. At the same time a stop-clock was started.
- The mixture in the beaker was stirred until a sudden colour change was seen.
- The stop-clock was immediately stopped and the time recorded.
- The beaker was rinsed with water.

Experiment 2

- Experiment 1 was repeated using 17 cm^3 of distilled water and 18 cm^3 of aqueous potassium iodate.

Experiment 3

- Experiment 1 was repeated using 21 cm^3 of distilled water and 14 cm^3 of aqueous potassium iodate.

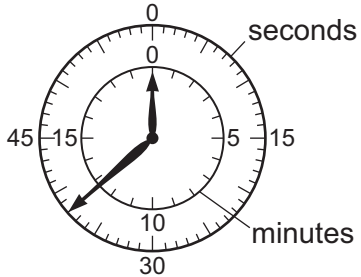
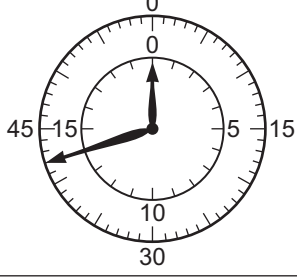
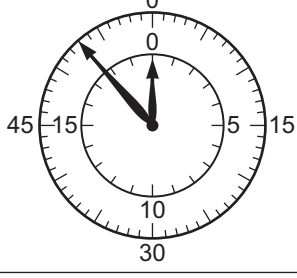
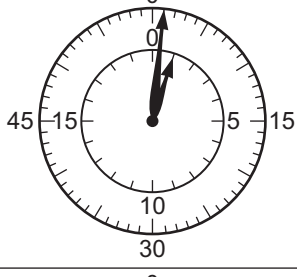
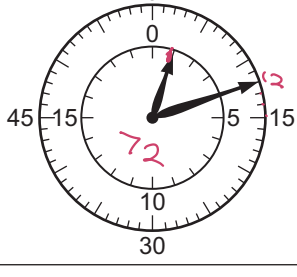
Experiment 4

- Experiment 1 was repeated using 23 cm^3 of distilled water and 12 cm^3 of aqueous potassium iodate.

Experiment 5

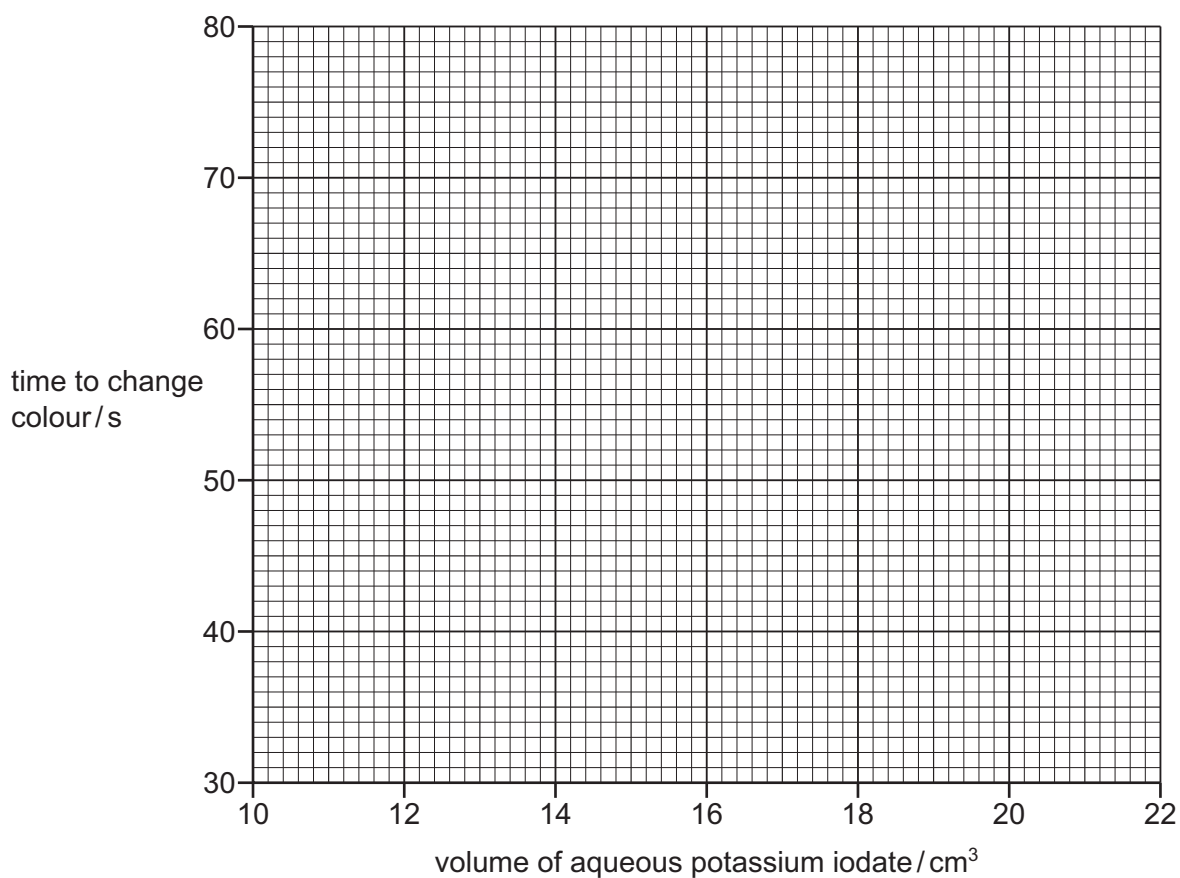
- Experiment 1 was repeated using 25 cm^3 of distilled water and 10 cm^3 of aqueous potassium iodate.

(a) Use the information in the description of the experiments and the stop-clock diagrams to complete the table. Record the times in **seconds**.

experiment	volume of aqueous sodium metabisulfite / cm ³	volume of distilled water / cm ³	volume of aqueous potassium iodate / cm ³	stop-clock diagram	time to change colour / s
1			20		
2			18		
3			14		
4			12		51
5			10		

[5]

- (b) Plot the results from Experiments 1 to 5 on the grid.
Draw a smooth curve of best fit.



[3]

- (c) (i) **From your graph**, predict the time to change colour if 16 cm^3 of aqueous potassium iodate was used.
Show clearly **on the grid** how you worked out your answer.

time to change colour = s [2]

- (ii) Calculate the volume of distilled water required if 16 cm^3 of aqueous potassium iodate was used.

volume of distilled water = cm^3 [1]

- (d) Sketch **on the grid** the graph you would expect if Experiments 1 to 5 were repeated at a higher temperature. [1]

- (e) The concentration of potassium iodate in the reaction mixture in each experiment can be calculated using the equation shown.

$$\text{concentration} = \frac{0.05 \times \text{volume of aqueous potassium iodate}}{45}$$

- (i) Calculate the concentration of potassium iodate in the reaction mixture in Experiment 2.

concentration = mol/dm³ [1]

- (ii) State which experiment, 1, 2, 3, 4 or 5, had the fastest rate of reaction.

..... [1]

- (f) Suggest why the volume of distilled water added to each experiment was increased as the volume of aqueous potassium iodate was decreased.

.....
 [1]

- (g) Give **one** change you could make to the apparatus used which would improve the results. Explain your answer.

change to apparatus

 explanation
 [2]

- (h) How could the reliability of the results of this investigation be checked?

.....
 [1]

[Total: 18]